

Ostergaard Holdings

Ostergaard Holdings had used one hurdle rate for eleven years, and over those eleven years the company had quietly turned into a different business.

Ostergaard made industrial instrumentation across three divisions. Filtration sold consumable cartridges and housings into pharmaceutical and food processing under multi-year contracts — steady, unexciting, and roughly 40 percent of revenue. Flow Control sold valves and actuators to the process industries. Test Systems built custom automated test equipment for semiconductor and automotive customers, sold project by project, with revenue that swung by 30 percent from year to year.

The capital committee applied a 9 percent hurdle rate to every project in every division. The rate had been set in 2015, and the memo that set it was still in the shared drive: cost of equity 10.2 percent, after-tax cost of debt 5.1 percent, weights taken from the balance sheet.

Two things had happened since.

The first was that the weights in the 2015 memo were book weights. Ostergaard carried \$460 million of debt against \$480 million of book equity, so the memo had used 49 percent debt. Ostergaard's market capitalization was now \$2,100 million. Nobody had rerun the calculation.

The second was that the divisions had drifted. Between 2015 and 2025, Filtration's share of capital employed had fallen from 46 percent to 31 percent, and Test Systems' had risen from 19 percent to 38 percent. No one had decided that. It was the accumulated result of eleven years of a single hurdle rate applied to projects with different risk: Test Systems proposals cleared 9 percent easily and Filtration proposals often did not.

Ingrid Solberg, newly appointed CFO, had two live proposals on her desk that made the pattern concrete. Filtration wanted \$70 million for a cartridge manufacturing line under a ten-year supply agreement with a pharmaceutical customer, returning 8.1 percent. Test Systems wanted \$70 million for capacity to serve an advanced-packaging customer, returning 10.4 percent, with a contract that ran two years and an option for a third.

Under the company hurdle rate, the answer was obvious. Solberg thought the answer was obviously wrong.

EXHIBIT 1 — OSTERGAARD CAPITAL STRUCTURE

Debt, book and market	\$460m
Book value of equity	\$480m
Market value of equity	\$2,100m
Pretax cost of debt	6.7%
Marginal tax rate	24%
Cost of equity, per the 2015 memo	10.24%
Risk-free rate	4.2%
Equity risk premium	5.0%

EXHIBIT 2 — DIVISIONAL RISK

DIVISION	UNLEVERED COMPARABLE BETA	TARGET DEBT/CAPITAL	SHARE OF CAPITAL EMPLOYED, 2015 ²⁰²⁵
Filtration	0.70	10%	46% 31%
Flow Control	1.10	18%	35% 31%
Test Systems	1.80	25%	19% 38%

EXHIBIT 3 — THE TWO PROPOSALS

	FILTRATION LINE	TEST SYSTEMS CAPACITY
Investment	\$70m	\$70m
Expected return	8.1%	10.4%
Contract length	10 years	2 years + option

ASSIGNMENT

- 1 Recompute Ostergaard's weighted average cost of capital using market weights. Compare to the 2015 memo. Which is right, and why is the wrong one wrong?

- 2 Compute a cost of capital for each division. Use the divisional betas and target capital structures given.
- 3 Apply the divisional rates to the two proposals. Which projects are accepted now, and which were accepted under the 9 percent rate?
- 4 Explain the 2015-to-2025 drift in the capital employed shares as a consequence of the hurdle rate policy. Was anyone responsible for it?
- 5 Solberg wants to move to divisional hurdle rates. Name the three objections she will hear from division heads and answer each one.